

Neural Processes Linking Interoception to Moral Preferences Aligned with Group Consensus

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Aligning one's decisions with the prevailing social norms and expectations of those around us constitutes a fundamental facet of moral decision-making. When faced with conflicting moral values, one adaptive approach is to rely on intuitive moral preference. Although theoretical accounts have proposed a link between moral preferences and interoceptive awareness—the capacity to sense internal bodily signals, this connection has not been empirically examined. This study examines the relationship between moral preferences and interoception, measured with self-report, heartbeat counting task, and resting-state functional magnetic resonance imaging. Two independent experiments demonstrate that both male and female participants' interoceptive awareness and accuracy are associated with their moral preferences aligned with group consensus. In addition, the fractional occupancies of resting-state brain states involving the ventromedial prefrontal cortex and the precuneus mediated this relationship. These findings provide empirical evidence of the neural mechanism linking interoception to moral preferences aligned with group consensus.

Key words: allostasis; heartbeat detection task; hidden Markov model; medial prefrontal cortex; precuneus; resting-state fMRI

Significance Statement

We investigate the relationship between interoceptive ability to perceive internal bodily signals and decision-making in moral dilemmas. Our findings demonstrate that greater interoceptive accuracy and awareness are associated with moral choices that align with group consensus. Additionally, resting-state brain activity in the ventromedial prefrontal cortex and precuneus mediates this relationship. These findings provide empirical support for the idea that internal bodily signals shape moral intuitions in ways that reflect social expectations across various social contexts.

Introduction

Moral intuition refers to the spontaneous, unconscious emergence of moral judgments with emotional weight, bypassing deliberate reasoning or evidence evaluation (Haidt, 2001). Moral decisions are strongly shaped by societal and cultural norms (Awad et al., 2018) and are driven by moral intuition to align with these norms and meet others' expectations (Haidt, 2001). Moral intuition can be particularly useful in moral dilemmas, which present choices that involve conflicting ethical values without clear right or wrong answers, but certain options are often more likely to be accepted and expected by the majority

within a social group (Greene et al., 2001; Koenigs et al., 2007). Recent theories propose that conforming to majority-approved moral preferences may be an efficient strategy to reduce metabolic demands, minimizing the effort required to infer others' expectations and adjust one's internal social model. This alignment reduces prediction error and increases the chance of one's survival (Constant et al., 2019; Theriault et al., 2021). Thus, individuals' moral preferences may reflect their internalized models of what the majority would expect in specific situations.

Moral intuition is closely linked to neural pathways involved in interoception (Woodward and Allman, 2007), the process by which the brain detects, interprets, and integrates internal bodily signals. Interoception plays a significant role in emotional processing and the sense of self (Craig, 2002; Critchley and Harrison, 2013; Barrett and Simmons, 2015; Khalsa and Lapidus, 2016). By integrating interoceptive signals, the brain assesses bodily states and external demands, predicting and regulating physiological adjustments (e.g., increasing heart rate, adjusting hormone secretion) to achieve allostasis (Sterling, 2012). This process helps build and maintain elaborate models

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of internal and external environments (Barrett et al., 2004; Farb et al., 2015), crucial for maintaining bodily homeostasis (Pezzulo et al., 2015). Interoceptive abilities are associated with individual differences in emotional perception and regulation (Wiens et al., 2010; Terasawa et al., 2013; Garfinkel et al., 2015), sophisticated and efficient decision-making (Dunn et al., 2010), and adaptive social cognition in context-dependent manners (Garfinkel and Critchley, 2013; Azevedo et al., 2017).

Interoceptive awareness-related brain activation has been reported in several regions, including the insular cortex, the anterior cingulate cortex (ACC), and the medial prefrontal cortex (mPFC). These regions have been identified as critical neural substrates for combining interoceptive information and subjective emotional experience. For example, activations in the insular cortex and ACC have been observed when participants direct their attention to their heartbeats (Critchley et al., 2004), and heartbeat-evoked potentials—electroencephalography signals observed 200 ms after the R-wave of the electrocardiography—have been found over the frontal and prefrontal cortex (Pollatos and Schandry, 2004).

The mPFC, especially its rostral subregion including the pregenual ACC, is considered a hub of the default mode network (Raichle et al., 2001) as well as the visceromotor cortical system (Barrett and Simmons, 2015; Farb et al., 2015; Stephan et al., 2016). The hierarchical structure of the mPFC is proposed to play a key role in the allostatic regulation of social decision-making (Smith et al., 2017b; Kim, 2020).

Recent findings also suggest that mPFC activity is linked to the process of adjusting one's choices to meet the expectations of others. For example, the mPFC has been shown particularly sensitive to social observation (D. Jung et al., 2018; Li et al., 2021; Yoon et al., 2021; Kim et al., 2023) and is involved in impression management and reputation-seeking behaviors (Yoder and Decety, 2018; Yoon et al., 2018; van Baar et al., 2019; Kim and Kim, 2021). Specifically, during resting periods, individuals with higher moral competence showed greater functional connectivity between the vmPFC and the amygdala (Jung et al., 2016). Stronger functional connectivity between the vmPFC and the precuneus during rest has also been linked to higher moral sensitivity in economic decision-making (Liu et al., 2020). Furthermore, individuals with stronger resting-state functional connectivity between the vmPFC and the precuneus exhibited higher levels of emotional awareness, a function closely tied to interoception (Smith et al., 2017a). Based on these findings, we predicted that the functional connectivity of neural circuits centered on the vmPFC would be associated with individual differences in moral intuition toward group consensus.

Recent neuroimaging studies have demonstrated that task-independent neural measures obtained from resting-state functional magnetic resonance imaging (fMRI) can be useful to predict specific behaviors such as financial decision-making (Kable and Levy, 2015; W. H. Jung et al., 2018) and social comparison (Jung and Kim, 2020) because they are less dependent on specific contexts and can be relatively stable over time. Hidden Markov models (HMMs) are a useful approach for characterizing the dynamics of brain states across time, as this approach describes brain activity as a dynamic sequence of discrete brain states with distinct patterns of network activity (Vidaurre et al., 2018, 2021). When applied to resting-state fMRI data, HMMs can capture quasistationary states of activity that consistently recur across a population (Baker et al., 2014), allowing for the investigation of how certain dynamic properties, such as differences in occupancy of brain states, vary across subjects.

In this study, we aimed to investigate whether interoceptive ability, measured by both subjective reports and a heartbeat detection task, is associated with the tendency of making moral decisions that align with group consensus, particularly in decisions involving moral dilemmas. Additionally, we examined if patterns of dynamic functional connectivity of the neural network centered around the mPFC, modeled using HMM on resting-state fMRI data, can predict individual differences in the degree to which moral preferences align with group consensus.

Materials and Methods

Participants

For Study 1, the participants, who were recruited through online communities of local universities in Seoul, were selected from a larger database where they initially participated in the data collection for another research project. These participants participated in a behavioral task before scanning and were scanned during rest to provide resting-state and structural images. Out of 109 participants recruited for the initial research project, 74 participants (mean age, 23.55 years; SD, 2.30; 36 males) had provided consent to participate in this study and provide their imaging data for this study, and were provided with an online link to the moral dilemma task and instruction on how to participate in the task. All participants performed the online moral dilemma task within a week after scanning. The same participants' resting-state functional MRI data were used to examine the neural signatures of moral decisions toward group consensus. Participants for Study 2 were recruited separately through online communities of local universities in Seoul. Thirty participants (mean age, 24.63; SD, 2.34; 15 males) were recruited and these participants participated in the moral dilemma task online, 1–2 d before visiting the laboratory to participate in the heartbeat counting task. The study design and the data collection procedures complied with all relevant ethical regulations and were approved by the Korea University Institutional Review Board.

Experimental design and statistical analyses

Moral dilemma task. Forty-eight moral dilemma scenarios used in the online moral dilemma task were selected from those used in two previous studies (Greene et al., 2001; FeldmanHall et al., 2014). These were translated into Korean first and rated for difficulty in vocabulary and coherency of the sentences by 14 independent raters who were all native Korean university students recruited through an online university community. Each rater read the scenario and rated the difficulty of the text on a five-point scale from 0 (not difficult at all) to 4 (very difficult). For the scenarios where the rating was above three points, the raters were asked which part of the scenario was difficult to understand and to provide an alternative sentence or words to make the scenario clearer to understand. The feedback from the raters was applied to modify and improve the translation of the scenarios used. The scenarios included 25 personal dilemmas, where participants had to decide whether to directly cause harm to another person for another moral cause, and 23 impersonal scenarios. The following is an example of a personal dilemma scenario: "You are a doctor. You are treating five patients who are nearing death due to organ failure. You are also responsible for one healthy young person. The only way to save the five patients with damaged organs is to transplant the organs of a healthy young man into the bodies of five people against his will. If that happens, the young man will lose his life, but he will be able to save five patients from death. Will you transplant the young man's organs into five patients to save them? (Greene et al., 2001)" In this example, a utilitarian choice would be to transplant the organs to save more people, whereas a deontological choice would be to choose to not to kill the healthy young man as killing would be wrong from a normative perspective.

The participants were asked to read the scenario and make a binary choice on how they would act in the scenario as the protagonist. The utilitarian choices were coded as 1 and the deontological choices were coded as 0. Then they were asked to rate the difficulty of their decision on a five-point Likert scale where 1 was extremely easy and 5 was extremely

difficult. The task was self-paced as each participant could have a different reading speed. The task was implemented using jsPsych library (de Leeuw, 2015), and the data were collected through the “Cognition.run” server.

Heartbeat counting task. In Study 2, interoceptive accuracy (IAcc) was assessed by a heartbeat counting task (Schandry, 1981). Participants were seated in front of a monitor and asked to concentrate on their internal body states and count their heartbeats from the start signal until the stop signal appeared on the screen. The start and stop signals were visually displayed on the monitor. The participants’ actual heartbeats were recorded using an electrocardiogram acquisition system (Biopac, MP36), with electrodes attached in the Lead 2 position. The task was conducted six times, each with a different duration (25, 30, 35, 40, 45, and 50 s). Participants were not informed of the length of each trial. The sequence of the trials was randomized once and fixed for all the participants. Participants were instructed to honestly report the number of heartbeats they felt without touching their body (e.g., wrist) to feel the actual pulse pressure. For each participant, individual IAcc was evaluated by the difference between the number of actual heartbeats and the reported heartbeats, calculated as follows (Pollatos and Schandry, 2008; Dunn et al., 2010):

$$\frac{1}{6} \sum_{n=1}^6 \left(1 - \frac{|\text{actual heartbeats} - \text{reported heartbeats}|}{\text{actual heartbeats}} \right).$$

Self-report questionnaires. All the participants completed the Korean multidimensional assessment of interoceptive awareness (K-MAIA; Gim et al., 2016), which is a 32-item Korean edition of MAIA (Mehling et al., 2012). This questionnaire comprises six subscales measuring awareness and attention to bodily sensations and the connection between body sensations and emotional states.

Behavioral data analysis. The group consensus moral decision for each scenario was first calculated as the ratio of participants who chose the utilitarian option. Then each participant’s moral distance from group consensus was calculated as the average Euclidean distance between the participant’s choice and the group-level ratio of utilitarian choice across all the scenarios. Moral similarity was defined as the percentage of an individual’s binary decisions matching the more preferred option at the group level.

The behavioral data were analyzed using the SPSS 25 software. One-sample *T* tests were performed to test whether the moral similarity to group consensus was above the chance level. Bivariate linear relationships between interoceptive awareness and moral similarity to group consensus and between interoceptive awareness and moral distance from group consensus were assessed by computing Pearson’s correlations. All reported correlation analysis results were significant at $p < 0.05$ level, corrected for multiple comparisons using Bonferroni’s method where more than two variables were considered.

Resting-state fMRI data acquisition and preprocessing. MRI data were acquired with a Siemens Magnetom Prisma 3 T scanner equipped with a 64-channel head coil. Whole-brain functional images during rest were acquired with the following parameters: TR, 1.35 s; TE, 0.03 s; slice thickness, 3 mm; imaging matrix, 80×80 mm; field of view, 240 mm; voxel size, $3 \times 3 \times 3$ mm; number of slices, 46; and flip angle, 90° . Three hundred four volumes of functional images were acquired for each participant and preprocessed following the default preprocessing pipeline in CONN Toolbox (20b) and SPM12. Images were first realigned and corrected for slice timing. Structural images were normalized to a Montreal Neurological Institute template with affine registration, followed by nonlinear transformation. Scans with excessive head motion-related artifacts were detected and discarded using the “intermediate” threshold (i.e., global bold signal changes, >5 standard deviation; framewise displacement, >0.9 mm). The normalization parameters determined for the structural volume were applied to the functional images. Images were then smoothed with an 8 mm Gaussian kernel at full-width at half-maximum. The resulting preprocessed resting-state fMRI data underwent additional postprocessing steps to reduce the artifact related to head motion and non-neuronal physiological variations. Regression of the (1) five principal components and temporal derivatives

from the white matter and cerebrospinal fluid time series using anatomical component-based noise correction procedure (Behzadi et al., 2007), (2) 12 realignment parameters, and (3) effect of condition (rest) and temporal bandpass filtering (0.008–0.09 Hz) were implemented simultaneously in a single regression step (Hallquist et al., 2013). Linear detrending was also applied.

Regions-of-interest selection. To investigate the sequence of discrete hidden brain states associated with social decision-making and interoception, we included nodes of canonical resting-state brain networks encompassing regions relevant to these processes. Specifically, we selected resting-state networks including the regions such as the anterior insula, mPFC, and temporoparietal junction from predefined and established networks using CONN v20.b. Our final regions of interest (ROIs) comprised 19 areas, including regions within Default Mode Network (DMN; MPFC, left LIP, right LIP, and PCC), Salience Network (SN; ACC, left anterior insula, right anterior insula, left RPFC, right RPFC, left SMG and right SMG), Dorsal Attention Network (DAN; left FEF, right FEF, left IPS, and right IPS), and Frontoparietal Network (FPN; left LPFC, right LPFC, left PPC, and right PPC).

Dynamic functional connectivity (HMM). For each participant and each of the 19 parcels, we extracted time series data by first calculating the mean signal, followed by demeaning and scaling based on its standard deviation. Subsequently, the individual time series for each participant were temporally concatenated to form a single matrix with dimensions (time, 304 volumes $\times$ 74 participants $\times$ 19 ROIs).

Next, an HMM was fitted to this concatenated time series. At the group level, we treated each participant’s time series as an independent sequence and fit a single HMM to the concatenated time series by using the “hmmlearn” package in Python. Each state in the HMM was defined by a multivariate Gaussian distribution, characterized by the mean within each voxel and covariance between voxels. The training was performed using 500 iteration cycles.

To determine the optimal number of brain states, we employed the Bayesian information criterion (BIC) score. Previous studies modeling hidden brain states in healthy participants suggested considering the number of states between 8 and 12 (Vidaurre et al., 2017, 2018; Kottaram et al., 2019). Consequently, we fit the HMM with 2–12 states and found that fitting it with 11 states yielded the best result (Fig. 3A). The model with 11 states was selected as the best model, and Viterbi sequence decoding was subsequently performed on this model. From the decoded temporal sequence, we calculated the fractional occupancy (FO) of each state for each participant.

Meta-analytic functional decoding. To quantitatively infer the function of each brain state, we conducted meta-analytic decoding on the whole-brain coactivation map for each state. For each participant and each brain state, we estimated a first-level GLM model with a single regressor representing the occupation of the brain state. Subsequently, one-sample *t* tests at the second level were performed for each brain state contrast to identify the whole-brain coactivation patterns associated with each state. The results were corrected using an FDR threshold of $q < 0.05$ to control for multiple comparisons.

Additionally, we generated topic-based meta-analytic maps using the NiMARE Python package and the latent Dirichlet allocation (LDA) model database, which utilizes features from the Neurosynth version 7 data release (github.com/neurosynth/neurosynth-data). The LDA model, applied to article abstracts from the Neurosynth database, identifies latent structures and provides semantically coherent neuroimaging topics. This approach serves as an alternative to term-based meta-analytic methods for functional decoding of brain networks (De La Vega et al., 2016; Wang et al., 2020). In our study, we utilized the LDA200 model, which models all abstracts in the Neurosynth database into 200 different, semantically coherent topics. We generated meta-analytic maps for each of the 200 topics in the following order: First, we labeled topic-related studies by applying a frequency threshold of 0.05 to the probability of a topic given an article, classifying them as topic-positive and the remaining studies as

topic-negative. Next, we fit the meta-estimator MKDACHi2 from the NiMARE package, resulting in 200 meta-analytic maps corresponding to each topic. Then, we calculated Pearson's correlations between each state's whole-brain coactivation map and the 200 meta-maps to identify the topic with the highest correlation. To intuitively understand the function of each state, we visualized the two most relevant words from each of the 10 topics that showed the highest correlation with each state, as illustrated in the word cloud (Fig. 3C).

Association between brain state dynamics and behavioral measures. We first identified the brain state related to moral decision-making tendency aligned to group norm by calculating bivariate correlations between FO of each state and moral distance. Multiple comparisons for correlation analyses were corrected using Bonferroni's method. Next, to assess the dynamics between brain states, correlations between FOs of the brain states were examined.

We tested whether the brain state showing heightened mPFC activation and related to the words such as evaluation and judgment, which we expected would be relevant in the moral decision-making process is associated with individual differences in interoceptive ability. For this purpose, the bivariate Pearson's correlation between FO of state 10 and the MAIA score was examined. Then, to identify the resting-state dynamic brain connectivity characteristics related to moral decision-making aligned to group norm, Pearson's correlation between all 11 state FOs and moral distance was examined, and significance levels were FDR-corrected. The moral distance measure was chosen over the similarity measure because it more accurately captures individual differences in moral decisions aligned with group norms. For instance, moral similarity—defined as the proportion of moral decisions matching the group consensus—treats scenarios with both high and low consensus equivalently. In contrast, when the group-level consensus is low, even a small binary decision difference can appear disproportionately significant, despite minimal actual deviation from the group consensus. To examine how the brain state dynamics related with the interoceptive ability and moral decision-making aligned to group consensus contributes to moral decision-making, we performed the mediation analysis using a bootstrapping approach with 5,000 samples. The model included the independent variable of interoceptive awareness measured with K-MAIA, the outcome variable of moral distance from moral dilemma task, and the mediating variable of the FO of States 8 and 10, which were the two brain states correlated with the moral distance and MAIA scores, respectively.

Results

Individuals' moral decisions resemble group consensus moral decision

The percentage of making moral choices that match the majority decision of all the participants across the study was assessed. The mean percentage of this moral similarity to group consensus was significantly above the chance level in Study 1 ($M = 0.790$; $SD = 0.054$; $t_{(73)} = 46.165$; Cohen's $d = 5.370$; $p < 0.001$) and in Study 2 ($M = 0.718$; $SD = 0.130$; $t_{(29)} = 9.177$; Cohen's $d = 1.677$; $p < 0.001$; Fig. 1A). This moral similarity to group consensus was not related to the average percentage of utilitarian decisions ($r_{(104)} = 0.016$; $p = 0.870$; Fig. 1B).

To examine whether the group consensus decisions in different scenarios are stable across different studies, we calculated the percentage of utilitarian choices for each scenario and assessed the correlation of these percentages across different participant samples from two studies. This analysis demonstrated that the group consensus percentages of the two independent study samples were highly correlated ($r_{(47)} = 0.968$; $p < 0.001$; Fig. 1C).

Interoceptive ability reflects moral alignment with group consensus

To test our *a priori* hypothesis, we assessed the relationship between participants' interoceptive awareness and moral decision aligned with group consensus or the degree to which they

make moral decisions in line with group consensus. The moral distance from group consensus was defined as the mean Euclidean distance of the participant's choice from the percentage of utilitarian decision at the group level for each scenario (i.e., moral distance from the group). For an example of the latter, if 30% of the group chose the utilitarian option (coded as 1) in a scenario and a participant chose the deontological option (coded as 0) for the scenario, then the Euclidean distance for that scenario would be $\sqrt{(0.3 - 0)^2} = 0.3$ for this participant, and if another participant chose the utilitarian option, the distance for that scenario of this participant would be $\sqrt{(0.3 - 1)^2} = 0.7$. This measure more accurately reflects the similarity of participant's decision to the group-level moral consensus than the percentage of making moral decisions matching group's majority decision, because for scenarios with lower level of consensus, such as scenarios with group-level utilitarian percentage near 50%, considering matching percentage would change drastically when assessing similarity in a binary manner. The moral distance calculated as above was negatively correlated with moral similarity, which is the participant's percentage of making binary choice matching the more preferred choice in each scenario across participants ($r_{(74)} = -0.826$; $p < 0.001$), indicating that this moral distance measure also reliably reflects how participant's moral decisions follows group consensus. Importantly, both indices of the group consensus moral decision were not related to the participant's percentage of making utilitarian decisions (moral similarity, $r_{(104)} = 0.016$; $p = 0.870$; moral distance, $r_{(104)} = -0.082$; $p = 0.410$).

In Study 1, participants' interoceptive awareness was measured using the K-MAIA (Gim et al., 2016), a self-report instrument assessing multiple dimensions of interoception. The bivariate relationships between the K-MAIA score and both moral similarity and moral distance were assessed. The moral distance measures were negatively correlated with the K-MAIA scores ($r_{(74)} = -0.240$; $p = 0.039$; Table 1). To investigate whether any specific attribute had a higher impact on the relationship, we conducted exploratory *post hoc* analyses to examine bivariate relationships between each MAIA subscale and moral similarity and moral distance. Among the subscales of MAIA, the moral distance showed a negative correlation with attention regulation subscale [$r_{(74)} = -0.243$; $p = 0.037$ (uncorrected); Table 1], which measures the ability to control and maintain attention directed to bodily sensations and constitutes a core concept of mindfulness (Gim et al., 2016). The K-MAIA score did not correlate with the ratio of utilitarian choices across all scenarios ($r_{(74)} = -0.178$; $p = 0.129$).

The bivariate relationships between K-MAIA score and moral similarity and moral distance for high- and low-level group consensus scenarios were assessed separately to assure that the relationship between interoceptive awareness and moral distance was not driven by the high consensus scenarios, where most participants easily agree on one of the two options. The K-MAIA score was positively correlated with moral similarity ($r = 0.369$; $p = 0.001$; Fig. 2A) and negatively with moral distance ($r = -0.354$; $p = 0.002$; Fig. 2B) in low consensus scenarios.

In Study 2, IAcc was measured using the heartbeat counting task (Schandry, 1981). Participants' IAcc score was positively correlated with moral similarity ($r_{(30)} = 0.422$; $p = 0.020$; Fig. 2C) and negatively correlated with moral distance ($r_{(30)} = -0.436$; $p = 0.016$; Fig. 2D). Again, participants' utilitarian choice ratios were not related to IAcc ($r_{(30)} = -0.039$; $p = 0.839$).

We recalculated the moral distance and similarity for each subject, excluding their decisions from the group norm

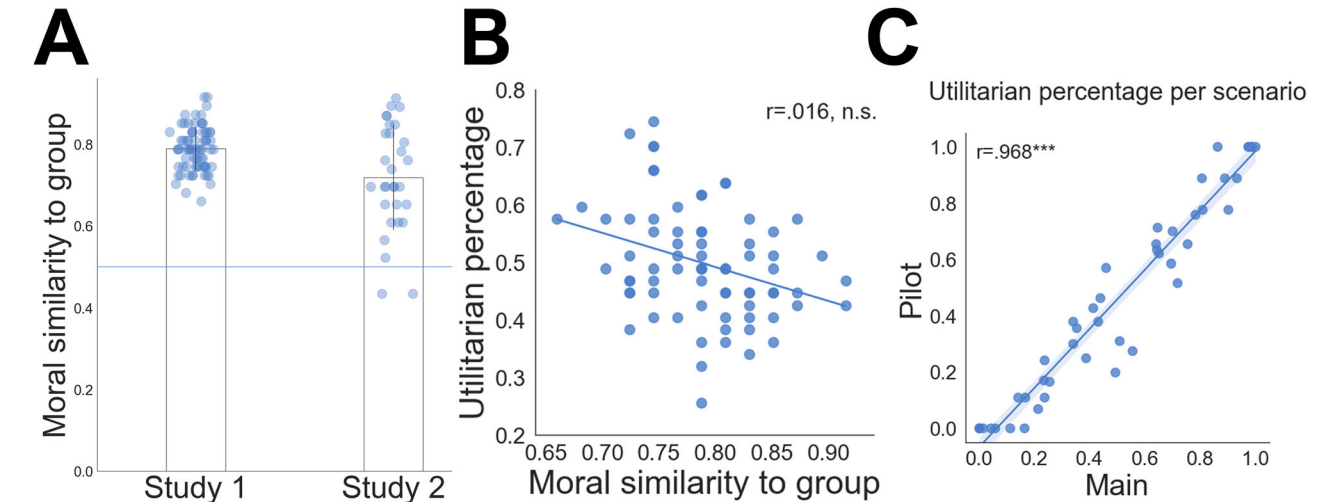


Figure 1. Quantifiable moral preferences aligned with group consensus in moral dilemma scenarios. **A**, Participants made moral decisions aligning with group consensus at above-chance levels in both studies. **B**, Moral similarity, defined as the percentage of decisions matching the group consensus, was not associated with participants' tendency to make utilitarian choices. **C**, The ratio of participants making utilitarian choices for each scenario was highly correlated across two studies, indicating consistency in group consensus decisions across experiments.

Table 1. Correlation between measures of moral decision aligned to group consensus and interoceptive awareness

MAIA subscale		Noticing	Accept	Attention regulation	Mind–body connection awareness	Return to body	Trusting	MAIA total
Moral distance	Pearson's <i>r</i>	−0.188	−0.085	−0.243 [†]	−0.131	−0.104	−0.144	−0.240*
	<i>p</i> value	0.108	0.473	0.037	0.267	0.377	0.222	0.039
Moral similarity	Pearson's <i>r</i>	0.144	0.173	0.138	0.114	0.229 [†]	0.071	0.237*
	<i>p</i> value	0.22	0.14	0.241	0.335	0.049	0.548	0.042

Moral distance, defined as the average distance between a participant's moral decisions and the group consensus across scenarios, was negatively correlated with the interoceptive awareness scale, suggesting that individuals whose moral decisions deviated further from the group consensus had lower interoceptive awareness. Among the subscales, the attention regulation scale, which measures the ability to control and maintain attention directed at bodily sensations, also showed a negative correlation with moral distance.

**p* < 0.05.
[†]*p* < 0.05 (uncorrected).

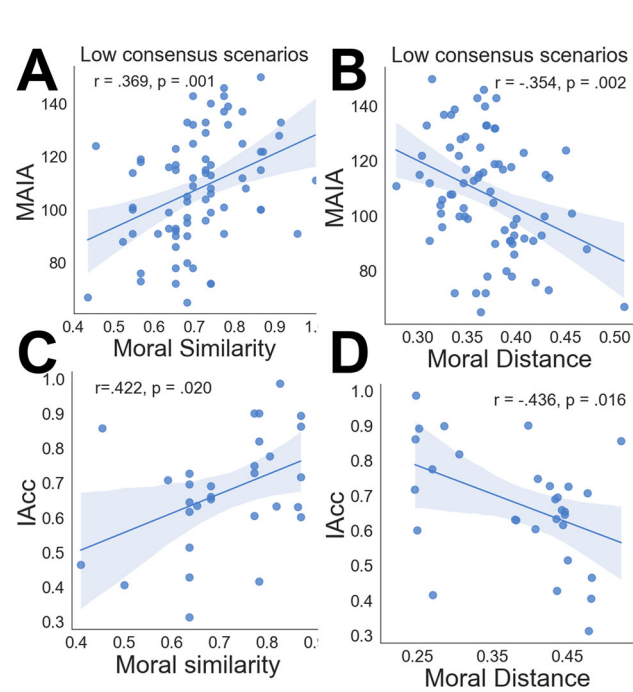


Figure 2. Correlation between moral decisions aligned with group consensus and interoceptive ability. Interoceptive awareness showed significant correlation with moral similarity (**A**) and moral distance (**B**) in scenarios with lower group consensus (Study 1). Additionally, moral similarity (**C**) and moral distance (**D**) were associated with IAcc as measured by the heartbeat counting task (Study 2).

calculation. These leave-one-out measures showed high correlation with the original values, and the relationships between moral distance/similarity and interoceptive awareness (Study 1) or IAcc (Study 2) remained significant (Table 2). We validated the results using the consensus from the other participant group (e.g., Study 2 group consensus for calculating distances and similarities for Study 1 participants and vice versa). The recalculated moral distance and similarity strongly correlated with the original measures based on the same group's moral norms, and all results remained significant (Table 3). In addition, we calculated the weighted moral distance by accounting for the degree of group consensus. For each scenario, the distance was adjusted by multiplying it by $1 - |c - 0.5|$, where c represents the group consensus (e.g., 80% consensus, $1 - |0.8 - 0.5|$). The adjusted distances were then averaged for each subject. In both Study 1 and Study 2, all results remained significant, and the weighted distances showed strong correlations with the original measures (Table 4).

Classifying hidden brain states during rest into diverse topics
The dynamic hidden brain states were analyzed using HMM applied to resting-state fMRI data. HMM allows for characterization of dynamic properties of spontaneous brain activity as this approach models the switching of brain states as a Markov chain, assuming that the observed brain activation pattern at each time point is generated from hidden states through a multivariate Gaussian distribution. The transition probability between the hidden states and the FO are often used to quantify the temporal dynamics of the inferred hidden states.

Table 2. Leave-one-out correlation between measures of moral decision aligned to group consensus and measures of interoceptive abilities

Study 1		Moral distance	Moral similarity	MAIA-Noticing	MAIA-Accept	MAIA-Attention regulation	MAIA-Mind–body connection awareness	MAIA-Return to body	MAIA-Trusting	MAIA total
Leave-one-out moral distance	Pearson's <i>r</i>	1.000**	−0.826**	−0.189	−0.085	−0.243†	−0.131	−0.104	−0.144	−0.240*
	<i>p</i> value	0.000	0.000	0.108	0.473	0.037	0.267	0.377	0.222	0.039
Leave-one-out moral similarity	Pearson's <i>r</i>	−0.817**	0.980**	0.126	0.135	0.16	0.135	0.261†	0.097	0.252*
	<i>p</i> value	0.000	0.000	0.283	0.25	0.173	0.252	0.025	0.412	0.03
Study 2		Moral distance	Moral similarity	HBC confidence	HBC accuracy					
Leave-one-out moral distance	Pearson's <i>r</i>	1.000**	−0.827**	−0.364	−0.438*					
	<i>p</i> value	0.000	0.000	0.052	0.015					
Leave-one-out moral similarity	Pearson's <i>r</i>	−0.883**	0.987**	0.345	0.441*					
	<i>p</i> value	0.000	0.000	0.067	0.015					

Moral distance and similarity were calculated for each subject by excluding their decisions from the group norm calculation. These leave-one-out measures were highly correlated with the original measures. Moreover, the relationship between moral distance/similarity and interoceptive awareness in Study 1, as well as IACC in Study 2, remained significant.
**p* < 0.05.
***p* < 0.01.
†*p* < 0.05 (uncorrected).

Table 3. Correlation between measures of moral decision aligned to group consensus and interoceptive awareness from independent samples

Study 1		Moral distance	Moral similarity	MAIA-Noticing	MAIA-Accept	MAIA-Attention regulation	MAIA-Mind–body connection awareness	MAIA-Return to body	MAIA-Trusting	MAIA total
Moral distance with Study 2 group norm	Pearson's <i>r</i>	0.876**	−0.793**	−0.198	−0.085	−0.277†	−0.174	−0.253†	−0.184	−0.322**
	<i>p</i> value	0.000	0.000	0.092	0.471	0.017	0.137	0.029	0.116	0.005
Moral similarity with Study 2 group norm	Pearson's <i>r</i>	−0.790**	0.807**	0.084	0.12	0.208	0.165	0.255†	0.173	0.273*
	<i>p</i> value	0.000	0.000	0.476	0.309	0.075	0.16	0.028	0.141	0.019
Study 2		Moral distance	Moral similarity	HBC confidence	HBC accuracy					
Moral distance with Study 1 group norm	Pearson's <i>r</i>	1.000**	−0.824**	−0.365	−0.439*					
	<i>p</i> value	0.000	0.000	0.051	0.015					
Moral similarity with Study 1 group norm	Pearson's <i>r</i>	−0.834**	1.000**	0.366	0.422*					
	<i>p</i> value	0.000	0.000	0.051	0.02					

Moral distance and moral similarity were calculated using group norms derived from participants in independent samples (e.g., calculating Study 1 measures with Study 2 group norms). The relationships between interoceptive abilities and the two measures remained significant.
**p* < 0.05.
***p* < 0.01.
†*p* < 0.05 (uncorrected).

Table 4. Correlation between measures of moral decision aligned to group consensus weighted by the degree of agreement for each scenario and interoceptive awareness

Study 1	Moral distance	Moral similarity	MAIA-Noticing	MAIA-Accept	MAIA-Attention regulation	MAIA-Mind-body connection awareness	MAIA-Return to body	MAIA-Trusting	MAIA total
Weighted distance	Pearson's r 0.984** 0.000	0.856** 0.000	-0.193 0.099	-0.129 0.273	-0.212 0.07	-0.147 0.212	-0.135 0.252	-0.139 0.238	-0.254* 0.029
Study 2	Moral distance	Moral similarity	HBC confidence	HBC accuracy					
Weighted distance	Pearson's r 0.939** 0.000	-0.834** 0.000	-0.372* 0.047	-0.468** 0.009					

Group norm was weighted according to the degree of consensus for each scenario. For each scenario, the moral distance was adjusted by multiplying it by $1 - |c - 0.5|$, where c represents group consensus (e.g., 80% consensus, $1 - |0.8 - 0.5|$). Adjusted distances were then averaged for each subject. In both Study 1 and Study 2, all results remained significant, and the weighted distances showed strong correlations with the original measures.
 * $p < 0.05$.
 ** $p < 0.01$.

We first obtained a group estimation of brain states on the concatenated time series data of all the participants from 19 ROIs, which yielded the model with 11 distinct brain states as optimal, based on comparison of BIC scores with differing number of brain states as the model parameter (Fig. 3A). The inversion of the optimal HMM yielded 11 distinct brain states (Fig. 3C), and Viterbi decoding yielded state time courses. Subject-specific FO, which denotes the proportion of time spent visiting each state, was calculated for each subject's time series. All states were significantly expressed at the group level (Fig. 3B).

We then performed functional decoding on the whole-brain coactivation maps of each brain state to understand the functional expression of the brain states from 200 topics. Each hidden brain state was associated with different groups of words (Fig. 3C).

Per our *a priori* hypothesis, we examined whether the dynamics of State 10, which was associated with the words such as evaluation and judgment and characterized by heightened activation in mPFC, was associated with interoceptive awareness. As predicted, the FO of State 10 was positively correlated with MAIA scores ($r_{(71)} = 0.247$; $p = 0.038$; Fig. 4D). No other states showed significant correlation with MAIA scores. Additionally, we assessed the relationships among brain states by examining correlations between their FO, which denotes the proportion of time spent in each state (Fig. 2D). The FO of State 10 was positively correlated with the FO of State 2 ($r = 0.31$; p -FDR = 0.035), which was associated with the words including "location" and "spatial" and characterized by lowered mPFC but heightened bilateral insula and dACC activation. It also showed a negative correlation with the FO of State 3 ($r = -0.039$; p -FDR = 0.005), which was associated with the words "memory," "retrieval," and "network" and characterized by higher mPFC and precuneus but lower bilateral insula activation, and with the FO of State 8 ($r = -0.45$; p -FDR = 0.0005), associated with the words "stimulation" and "pain" and characterized by overall lowered whole-brain activation with especially lowered precuneus activation. This association between State 10 and MAIA score was also present in an independent dataset, where both MAIA scores and resting-state fMRI data were collected (Extended Data Text S1). We analyzed the dynamic connectivity change in an independent dataset and functionally decoded each state using meta-analysis maps as done in the original dataset. This dataset was best explained with nine hidden states (Extended Data Fig. 1-1), from which we identified the state most similar to State 10 (Extended Data Fig. 2-1). The dynamic characteristics of this state was also associated with the MAIA score (Extended Data Fig. 3-1).

Hidden brain dynamics related to moral preferences aligned with group consensus

Our primary focus was on the neural dynamics underlying individual tendencies in moral decision-making aligned with group norms. Thus, we assessed the neural dynamics associated with moral distance from group consensus, as measured during the moral dilemma task, by analyzing bivariate correlations between moral distance and the FOs of each brain state. Moral distance from group consensus was positively correlated with the FO of State 8 ($r_{(74)} = 0.329$; p -FDR = 0.04404). However, contrary to our expectations, the FO of State 8 was not significantly associated with interoceptive awareness ($r_{(74)} = 0.074$; $p = 0.530$).

To further explore the neural dynamics of State 8, we examined correlations among the FOs of different brain states. The

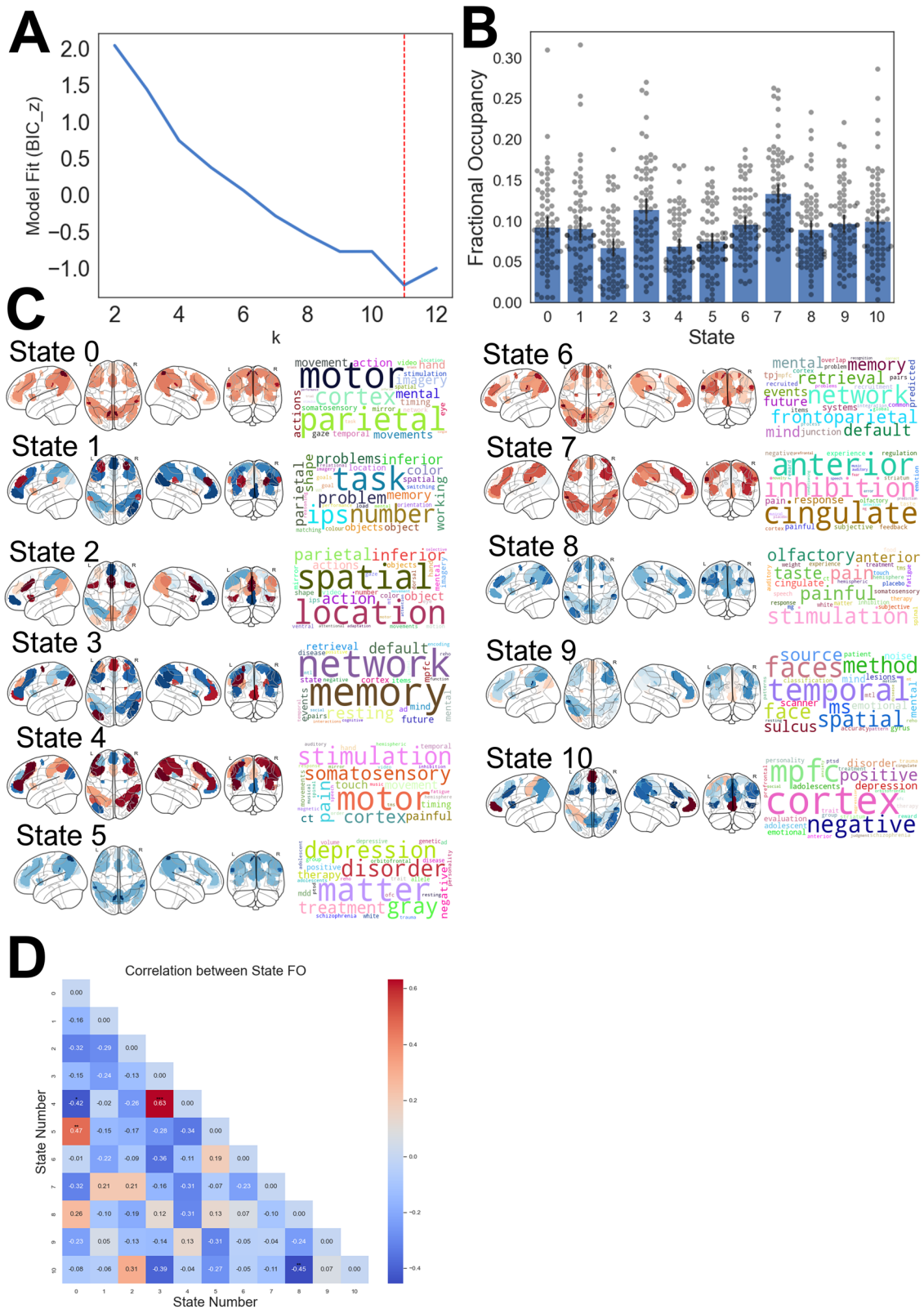


Figure 3. Brain states inferred with HMMs. **A**, An 11-state model were selected as the optimal model based on BIC scores. **B**, Each brain state was significantly visited at the group level, with varying individual differences in FO. **C**, State loading maps and word clouds for each brain state. The word size represents the correlation value between the topic of origin and the state. If a word appeared in multiple topics, the correlation values were summed. **D**, Correlations between state FOs.

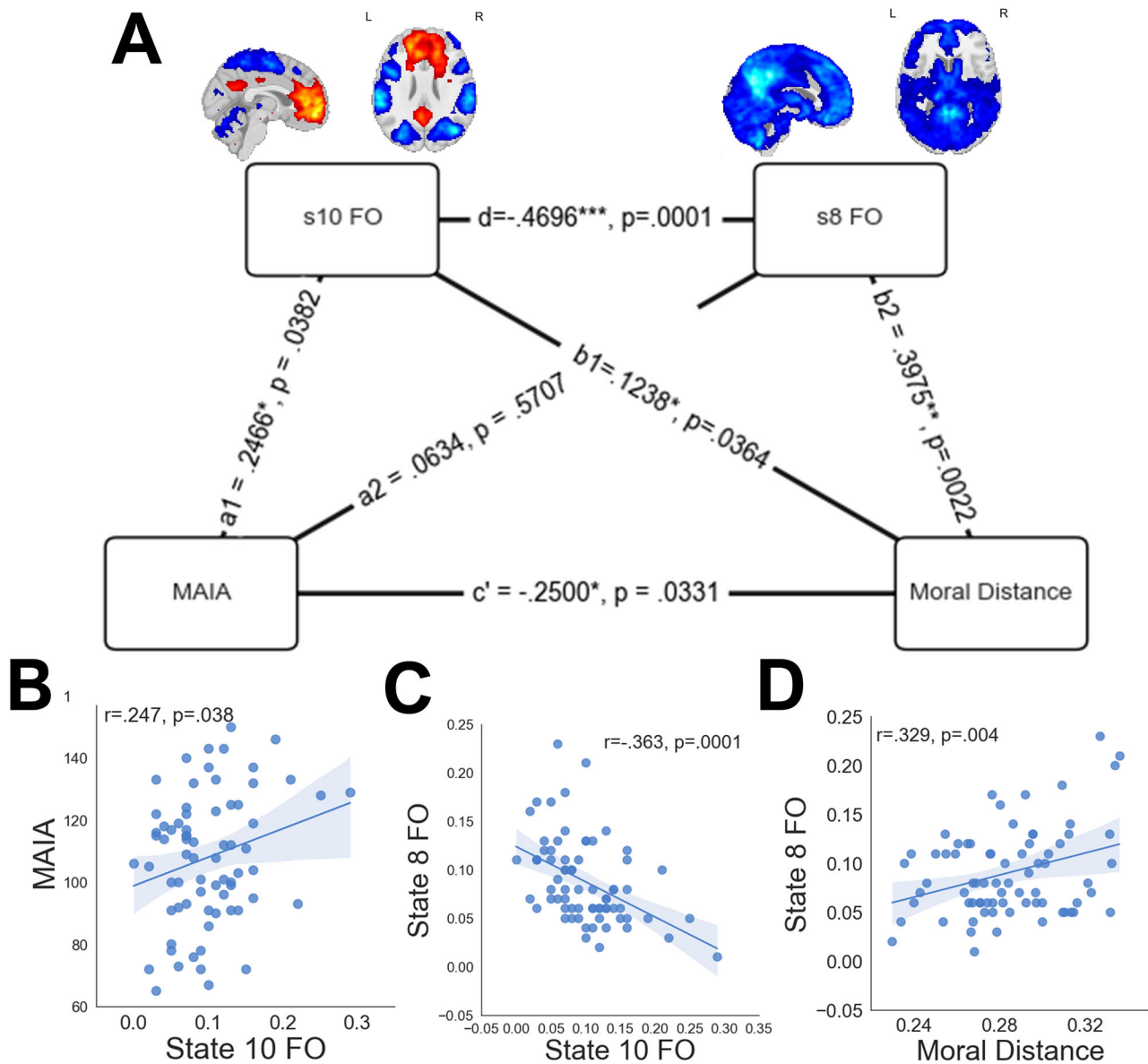


Figure 4. A dual mediation model testing the indirect effect of State 10 and State 8 FOs on the relationship between interoceptive awareness and moral distance. **A**, Interoceptive awareness indirectly influenced moral distance through the FOs of State 10 and State 8. **B**, The total effect of interoceptive awareness on moral distance showed a weak trend. **C**, State 10 and State 8 FOs were significantly correlated. **D**, Moral distance was significantly correlated with State 8 FO.

FO of State 10 was negatively correlated with the FO of State 8 ($r_{(74)} = -0.454$; $p\text{-FDR} = 0.008$), while no other brain state showed a significant relationship with State 8.

As we expected the dynamic features of State 10 to be related to moral distance, we sought to explore the underlying mechanisms between interoceptive awareness and moral distance through a dual mediation model involving FO of States 8 and 10 as mediators. This dual mediation model was tested based on the hypothesis that interoceptive awareness would influence moral preferences aligned with group consensus indirectly through a participant's time spent on the state associated with valuation during rest and whether this neural dynamic characteristic is mediated through another dynamic feature (i.e., FO of State 8).

The analysis revealed that the total effect of MAIA scores on moral distance was not statistically significant, though a weak trend existed ($b = -0.0003$; $p = 0.0895$), indicating a lack of direct association between subjectively reported interoceptive awareness and moral distance from the behavioral task. However,

examination of the indirect effects yielded significant findings. First, the indirect effect of MAIA on moral distance through FO of State 10 was significant (effect, 0.0678; BootSE, 0.0468; 95% CI [0.0024, 0.1759]), suggesting that FO of State 10 mediates the relationship between the MAIA and moral distance. Also, the indirect effect of both FO of State 10 and State 8 was significant (effect, -0.0460 ; BootSE, 0.0266; 95% CI [$-0.1059, -0.0028$]). However, the indirect effect of FO of State 8 alone was not significant (effect, 0.0252; SE, 0.0497; 95% CI [0.0817, 0.1199]). The direct effect of MAIA on moral distance was significant (effect, -0.0003 ; SE, 0.0001; $p = 0.0331$; Fig. 4).

Discussion

This study explored the potential link between interoception and an individual's inclination to align their moral preferences with group consensus, the moral choices accepted by the majority. As predicted, interoceptive awareness was positively correlated with

the extent to which participants made moral decisions in line with group consensus across various scenarios. Furthermore, resting-state brain dynamics among functionally distinct brain states were associated with this tendency. Specifically, there was a significant indirect effect of a brain state characterized by heightened activity in the mPFC on moral distance, mediated by a brain state characterized by lowered activity, especially in the precuneus. These findings underscore interoception as a potential biological foundation for constructing accurate internal models of social expectations and leading to moral preferences aligned with group consensus—an efficient strategy for minimizing metabolic costs related to bodily maintenance.

Humans both influence and are influenced by their social surroundings. Empirical evidence from various domains, such as public health campaigns (Falk et al., 2015), crowdfunding projects (Genevsky et al., 2017), judgments of facial preference (Kim et al., 2007), and assessments of facial trustworthiness (Engell et al., 2007), supports the idea that our brains track the consensus of those around us across a wide range of contexts. This study extends this body of research into the domain of moral decision-making, bridging the gap between empirical findings and theoretical explanations of consensus-based decision-making. It provides empirical evidence for the existence of group consensus in moral preferences, even in situations without universally “right” answers. Notably, this study elucidates that individuals typically make moral choices aligning with the prevailing group consensus, and it highlights the role of interoceptive awareness in shaping these individual differences.

Individuals develop internal models reflecting societal norms and others’ expectations within their social environment (Constant et al., 2019; Theriault et al., 2021). These models form one’s “sense of should” (Theriault et al., 2021), which drives individuals to align their behavior with cultural or social norms (Ramstead et al., 2016; Constant et al., 2019). Adhering to this “sense of should” and conforming to others’ expectations contribute to maintaining a more predictable social environment, which in turn conserve metabolic resources by reducing the cognitive and physiological costs of navigating complex social interactions (Theriault et al., 2021). In this study, participants were unaware of group norms, faced no social pressure, and made subjective judgments based on internalized moral criteria formed through life experiences. These criteria, shaped by repeated social interactions, may manifest as moral intuitions—a mechanism evolved to conserve bodily resources by aligning with others’ expectations. Theriault et al. (2021) propose that this “sense of should” operates to minimize resource expenditure by motivating norm-consistent behavior. Notably, the tendency to align with group preferences in moral judgments observed in this study is distinct from traditional moral conformity, as participants were not exposed to any external preference cues and made decisions based solely on their own internal moral preferences. Indeed, recent research found that higher interoceptive sensitivity is associated with reduced conformity (von Mohr et al., 2023), indicating that individuals with heightened bodily awareness may be more resistant to external influences that conflict with deeply internalized intuitions shaped through a lifetime of social experience.

According to recent research, interoception is associated not only with moral preferences but also with prosociality (Contreras-Huerta et al., 2023), indicating that it may serve as a common underlying mechanism across a broad range of social behaviors. In that study, participants made their choices after viewing explicit information about the benefits to themselves or others,

whereas in the present study, participants made moral decisions without any information regarding others’ moral preferences or decisions. This difference may account for distinct neural correlates observed: while dorsal ACC activity was linked to prosociality in the former study, ventromedial PFC activity was associated with moral preferences in the present one.

Findings show consistent moral preferences across different participant groups despite varied scenarios. Importantly, individuals whose moral decisions aligned with group preference tended to exhibit higher’ interoceptive sensitivity, suggesting a biological link between sensitivity to bodily signals and the motivation to align with group values. The ability to predict others’ expectations may depend on sensitivity to discrepancies between predicted and actual social outcomes—whether in behavior’ or emotional response. Those more attuned to such mismatches may refine their predictive models through bodily signals indicating homeostatic imbalance, which may facilitate the development of accurate social expectations.

Interoceptive signals convey essential information about an organism’s internal state for maintaining homeostasis (Craig, 2002), enabling adaptive responses to a dynamic environment by minimizing discrepancies between predicted and actual bodily states (Seth, 2013; Barrett and Simmons, 2015). The present study supports this theoretical framework by demonstrating that the inclination to align moral decisions with group consensus is associated with interoceptive awareness. Interoceptive awareness reflects one’s ability to attend to bodily signals and appropriately weigh them in shaping perception during a heart-beat counting task (Ainley et al., 2016). This optimization process enables individuals to generate prediction errors and effectively update their generative model. Consequently, individuals with better interoceptive ability can more precisely identify and allocate appropriate weight to relevant information sources when interpreting physiological responses in diverse social contexts (Ainley et al., 2016), leading to more refined internal models of the social environment.

It is important to note that interoceptive sensitivity does not correlate with the consistency of an individual’s utilitarian or deontological decision-making, consistent with recent research (Tamura et al., 2022). Previous studies have reported associations between certain neural structures, particularly the vmPFC, a critical component of the interoceptive neural circuitry, and the ratio of deontological versus utilitarian decisions (Koenigs et al., 2007). However, since individuals typically anticipate deontological choices to be socially approved in moral dilemmas, the observed shift toward utilitarian decisions following vmPFC lesions may reflect an impaired ability to accurately infer societal expectations in morally complex situations. Supporting this view, both behavioral impairment following vmPFC lesions and heightened vmPFC activity were primarily evident in personal moral scenarios, where deontological choices—such as avoiding physical harm to a minority—typically align with the broader population’s expectations (Lee et al., 2014, 2018). To avoid this confounding effect, our study carefully selected moral scenarios to eliminate potential correlations between group consensus and utilitarian (or deontological) decisions, shedding light on the need for further research to elucidate the precise role of the vmPFC in moral dilemma.

We investigated whether hidden brain state dynamics during rest are associated with participants’ inclination to make moral decisions aligned with group consensus, using a data-driven approach. This study modeled the hidden brain dynamics of the core nodes within well-established canonical functional brain

networks consistently engaged in social cognition, decision-making, and allocation of attention to internal and external information—specifically, the FPN, SN, DMN, and DAN. The inferred brain states were then functionally decoded by utilizing a large neuroscience research database, without preselecting relevant topics, to assure that all possible functional topics were considered. Our *a priori* hypothesis proposed that brain states characterized by heightened vmPFC activity would act as a hub within the neural network, linking interoceptive awareness to moral intuition aligned with group consensus. To test this, we examined whether State 10 met these criteria and found that it demonstrated heightened vmPFC activity and a significant correlation with interoceptive awareness, but not with moral distance. Further mediation analysis revealed that State 10 was indirectly associated with moral distance through its connection with State 8. The MPFC and the precuneus (or the posterior cingulate cortex) are major components of the DMN (Buckner et al., 2008; Babo-Rebelo et al., 2016) implicated in self-related processing (Buckner et al., 2008; Qin and Northoff, 2011) and monitoring of bodily signals (Thayer et al., 2012). Our data revealed that prolonged occupancy of a brain state with significantly lowered precuneus activity is associated with the extent to which a participant's moral decisions deviated from the general consensus. The precuneus serves various complex cognitive functions, including autobiographical memory retrieval, self-related information processing, and integration of environment-related perceptual information (Lou et al., 2004; Lyu et al., 2021) and is a major hub within the DMN (Utevsky et al., 2014) potentially playing a pivotal role in social and moral decision-making (Baumgartner et al., 2013; Kim and Kim, 2021). The indirect relationship between brain dynamics and the tendency that align moral decisions with group consensus could serve as a crucial biological foundation for moral intuition, supporting the development and use of internal models that accurately reflect others' expectations in complex social environments.

Some limitations of this study should be noted. First, while resting-state functional connectivity reveals stable individual traits over time and is a valuable tool for understanding context-independent neural characteristics and predicting behaviors, task-based fMRI may provide a more precise understanding of the mechanisms directly involved in interoceptively driven moral decision-making. Second, there is evidence that heightened interoceptive sensibility is associated with maladaptive mental health conditions. For example, individuals with high levels of interoceptive sensibility are more likely to experience panic disorders (Ehlers and Breuer, 1992; Yoris et al., 2015), anxiety disorders (Pollatos et al., 2007), and depression (Paulus and Stein, 2010). While the effort to predict and meet others' expectations is generally an adaptive trait, excessive efforts can become maladaptive, a phenomenon that may be explained by allostatic overload (Kim, 2020; Bobba-Alves et al., 2022). Identifying the dispositional and situational factors that determine whether interoceptive sensibility functions as an adaptive or maladaptive trait in social contexts represents an important direction for future research. Third, the group consensus in moral dilemma scenarios could be culturally specific. Although we confirmed consistent patterns of group consensus across independent participant samples, both were drawn from Korean university student populations. Possible cultural differences may exist, and future studies could help identify culture-specific versus universal moral norms, as well as their underlying neural circuits and mechanisms, through cross-cultural comparisons.

In conclusion, this study provides empirical evidence that interoception plays a significant role in shaping moral preferences aligned with group consensus. Importantly, activity in the mPFC and the precuneus, key regions of the default mode network involved in monitoring internal states at rest, was found to mediate this relationship. These findings provide scientific support for the notion that human moral intuitions emerge through the internalization of preferences that balance bodily needs with social expectations in the context of interpersonal interactions.

Data availability

The data for this study are publicly accessible at https://osf.io/432nf/?view_only=f0b4cc345c664493adddfec8aea1486d9. The code used for inferring HMM is publicly available at <https://hmmlearn.readthedocs.io/en/latest/>.

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